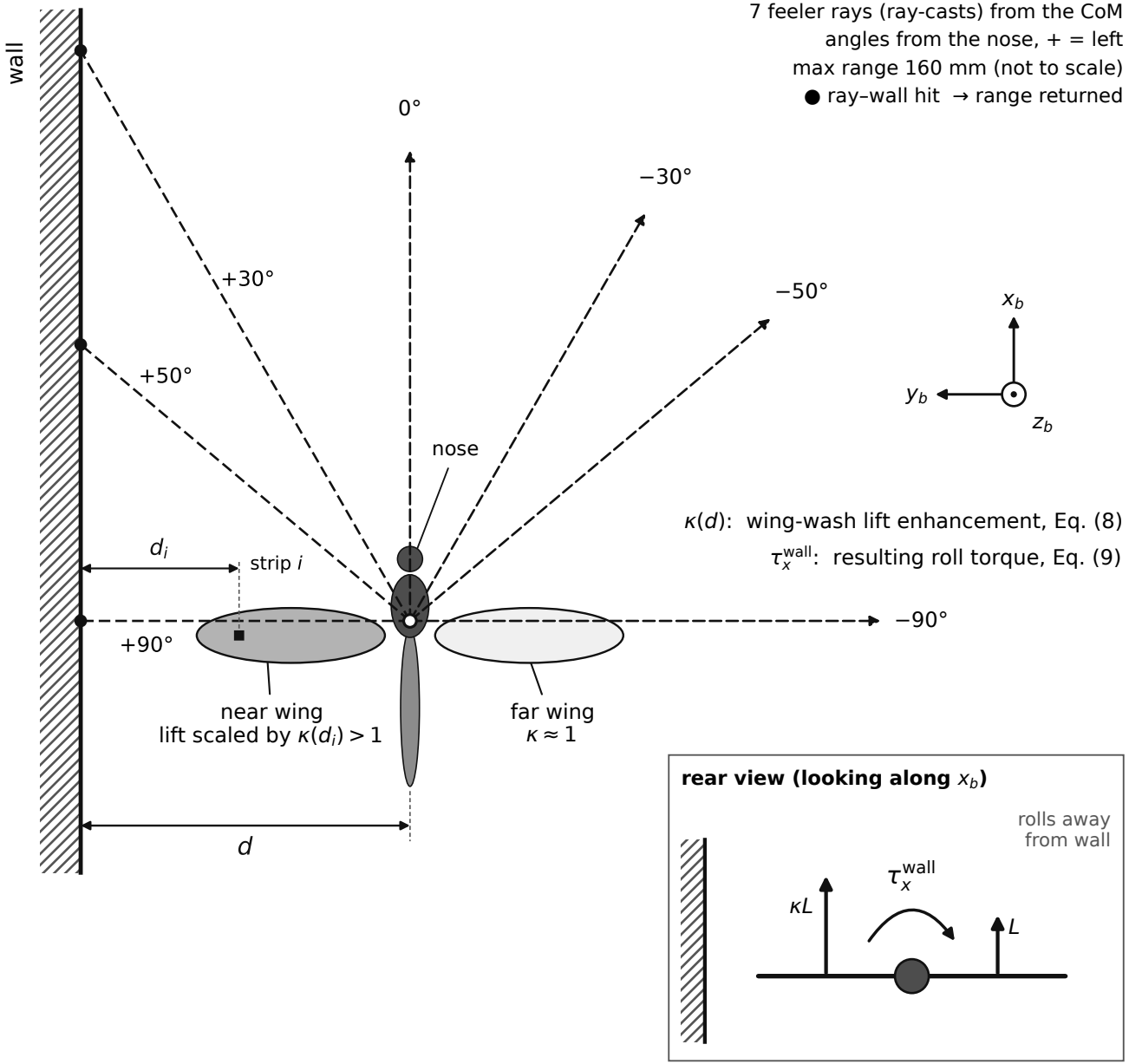
7 feeler rays (ray-casts) from the CoM
angles from the nose, + = left
max range 160 mm (not to scale)
● ray–wall hit → range returned
wall
0°
−30°
+30°
−50°
+50°
x_b
y_b
z_b
nose
$\kappa(d)$: wing-wash lift enhancement, Eq. (8)
τ_x^{wall}: resulting roll torque, Eq. (9)
d_i
strip i
+90°
−90°
near wing
lift scaled by $\kappa(d_i) > 1$
far wing
$\kappa \approx 1$
d
rear view (looking along x_b)
rolls away
from wall
κL
τ_x^{wall}
L